Image Collection

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for our PPT we need to start collecting **good context/reference images** related to the different directions we're considering for the project.

Our umbrella topic is:

“AI-Based Computer Vision for Automated Inspection Using Deep Learning”

Our current proposed application is **Earth observation / environmental analysis using satellite imagery**, but we want the PPT to show the broader possibilities of the technology as well.

1. Satellite / Earth Observation

Find good high-quality visuals of:

• Earth viewed from space

• Sentinel-2 / Earth-observation satellite

• Satellite orbiting Earth and taking images

• Satellite sensor/imaging payload

• Satellite observation swath / scanning the Earth's surface

• Multispectral satellite imagery concept

• Satellite imagery being converted into maps/data

Prefer actual Sentinel-2 / Copernicus-style imagery or scientifically accurate illustrations rather than generic “AI satellite” graphics.

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2. Example Satellite Images

We need sample satellite images showing different environments:

• Dense forest

• Agricultural fields

• Urban / concrete expansion

• Rivers / lakes

• Coastlines

• Desert / arid land

• Mountains

• Grasslands

• Wetlands

• Industrial areas

• Landfills / waste sites if good imagery exists

Ideally find **before/after examples from the same geographic location** where possible.

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3. Same Location — Different Time

This is important for our proposed **change-detection** idea.

Find examples where:

Same location + different dates

For example:

```text

`Location X — 2018`

`↓`

`Location X — 2024`
```

with visible changes such as:

- Forest → cleared land
- Vegetation → urban development
- Agriculture → buildings
- Water body → reduced water
- Coastline changes
- Mining expansion
- New roads
- Landfill expansion
- Deforestation
- Urban sprawl

If possible, find images where the two images are already properly aligned/registered.

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4. Image Registration / Alignment

We also want to demonstrate the computer-vision problem:

******Two images of the same place aren't necessarily perfectly aligned.******

Find visual examples of:

````text`

`Image A`

`+`

`Image B`

`↓`

`Image Registration`

`↓`
````

```
`Aligned Images`

`↓`

`Difference / Change Map`
```

Useful concepts to search for:

• satellite image registration

• remote sensing image co-registration

• image alignment

• feature matching

• homography

• geometric registration

• satellite change detection

A good visual would show two slightly misaligned satellite images and then the correctly aligned result.

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5. Change Detection

Find examples of AI/computer-vision change detection from satellite imagery.

Especially:

• Before / after satellite imagery

• Difference maps

• Semantic change detection

• Deforestation detection

• Urban expansion detection

• Flood detection

• Water-body change

• Land-use change

Ideally examples where the output is something like:

```text
BEFORE → AFTER → CHANGE MAP
```

```
rather than just two photographs.

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### 6. Land-Cover Segmentation

We need examples showing how AI converts a satellite image into a ******semantic segmentation map******.

For example:
```

````text
`Original satellite image`

`↓`

`AI / Deep Learning`

`↓`

`Forest = one class`
`Water = one class`
`Urban = one class`
`Agriculture = one class`
`Bare land = one class`
````

Search for:

• satellite semantic segmentation

• Sentinel-2 land-cover segmentation

• land-use land-cover classification

• satellite image segmentation

• U-Net satellite segmentation

• remote sensing semantic segmentation

A visual with the original image next to its segmentation mask would be ideal.

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7. Multispectral Imaging

We should have at least one visual explaining that Sentinel-2 isn't just an RGB camera.

Find visuals showing:

```text
`Blue`
`Green`
`Red`
`Red Edge`
`NIR`
`SWIR`
`...`
```

```
and examples of:
```

- RGB composite
- False-colour composite
- NIR imagery
- NDVI / vegetation map

Something that visually explains:

******“Same location, different spectral bands → different information.”******

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8. Ecological Condition / Environmental Analysis

Find visual examples of satellite-based environmental indicators:

- Vegetation health
- NDVI maps
- Forest cover
- Water availability
- Soil/bare-land maps
- Built-up area
- Surface temperature
- Ecological condition indices
- Environmental stress maps

We don't necessarily need to use these exact metrics; these are for explaining what our system could eventually derive from satellite imagery.

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9. Broader Computer Vision Applications

Since our original topic is deliberately broad, we also want a few context images showing other possible applications of ******AI + CV + Deep Learning for automated inspection******.

Potential areas:

******Manufacturing******

- Automated defect inspection
- Missing-component detection
- Assembly verification
- Surface defect detection

******Infrastructure******

- Road/pothole inspection
- Railway inspection
- Bridge/crack detection
- Structural damage detection

******Safety******

- PPE detection
- Industrial safety monitoring
- Fire/smoke detection
- Crowd anomaly detection

******Agriculture******

- Crop disease detection
- Plant health
- Weed detection
- Yield/environmental monitoring

******Medical******

- Blood-cell analysis
- Histopathology
- Medical image segmentation

******Transportation******

- Traffic analysis

- Vehicle detection
- Road condition
- Driver monitoring

******Satellite / Earth******

- Land-cover classification
- Deforestation
- Urban expansion
- Flood detection
- Marine debris
- Environmental monitoring

These are mainly for showing our professor that ******“automated inspection using computer vision” is a huge umbrella******, and we're narrowing it toward Earth observation.

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10. The Big Earth-OSINT Vision

For the final part of the PPT, we'd like some visuals representing the long-term idea:

******Open Earth Observation Intelligence******

Something like:

````text
`Satellite Data`
`+`
`Historical Imagery`
`+`
`Maps`
`+`
`Environmental Data`
`+`
`Public/OSINT Information`
`↓`
````

```
`AI`

`↓`

`Environmental Intelligence`
```

Find visuals of:

• Interactive Earth map

• Satellite imagery dashboard

• Geospatial intelligence interface

• Environmental monitoring dashboard

• Earth observation platforms

• Historical satellite timeline

• Global environmental maps

• AI analysis of satellite imagery

The eventual idea is a platform where a user could select a location and ask:

> “What is happening to this area?”

and the system could combine satellite imagery, historical imagery, computer-vision analysis and eventually other open sources to produce evidence-based environmental information.

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11. Very Important: Source Quality

Whenever possible, prioritize images from:

• NASA

• ESA

• Copernicus

• Sentinel / Sentinel Hub

• USGS

• Google Earth Engine documentation/examples

• Research papers

• IEEE / Springer / ScienceDirect papers

• University research labs

• Official government Earth-observation portals

For research-paper figures, record:

Paper title + year + figure number + source/link

For satellite imagery, record:

Location + date + satellite/sensor + source

Don't just download random Pinterest/Google images because we may need to cite them later.

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12. What I want at the end

Ideally make a folder structure like:

```text
PROJECT_VISUALS/
```

`01_Earth_Satellite/`

`02_Sentinel2/`

`03_Multispectral/`

`04_Land_Cover/`

`05_Image_Registration/`

`06_Change_Detection/`

`07_Ecological_Analysis/`

`08_Deforestation/`

`09_Urban_Expansion/`

`10_Water_Flooding/`

`11_Manufacturing/`

`12_Infrastructure/`

`13_Agriculture/`

`14_Surveillance/`

`15_Earth_OSINT/`

`16_Dashboards/`

`17_Papers_Figures/`

```
And for the ******most important visuals******, save both:

``image.jpg``

and

``source.txt``

containing the original source/link and a one-line description.

The most important images for our PPT are probably:

******1. Satellite orbiting Earth******

******2. Real Sentinel-2 image******

******3. Multispectral bands / false-colour image******

******4. Satellite image → AI → segmentation map******

******5. Same location at two different dates******
```

******6. Before → after → change detection map******

******7. Image registration/alignment******

******8. NDVI / ecological-condition map******

******9. Earth observation dashboard******

******10. Big-picture Earth + AI / environmental intelligence visual******

Don't worry about finding the exact final images yet. We mainly need a ******strong visual library****** from which we can choose when making the PPT.